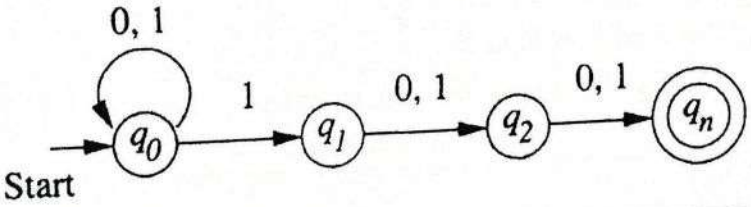
NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science & Engineering
END SEMESTER EXAMINATION, Jan-June 2025

B. Tech. (UG) Semester- 04
Course Name: Automata Theory & Compiler Design Code: CS44114
Maximum Time: 3 hours Max. Marks: 60

		Marks	CO	BL
1.	a) Assess whether the following CFG are LL(1) or not (provide proper justification): i) $S \rightarrow aAB \mid BAa$ $A \rightarrow aA \mid b$ $B \rightarrow aB \mid c$ ii) $S \rightarrow aA \mid b$ $A \rightarrow bA \mid a$ b) Convert the following NFA to equivalent DFA using subset construction method: 	5+5	CO1	A
2.	Convert the given grammar to LL(1). Prove that the grammar is really converted to LL(1) $S \rightarrow Sa \mid Sab \mid cA$ $A \rightarrow Ad \mid a$	5+5	CO2, CO3	E
3.	Construct LL(1) parse table for the following grammar and parse the string $aabbbcc$ $T \rightarrow R$ $T \rightarrow aTc$ $R \rightarrow bR \mid \epsilon$	6+4	CO2, CO3	C
4.	Construct LR(0) parse table for the following grammar and parse the string $id * (id + id)$ $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow id$	6+4	CO2, CO3	

5.	Write the formal definition of push down automata and its application. Design a pushdown automata to recognize the following language: $L = \{0^{2n}1^n : n \geq 0\}$ **OR** Write semantic rules for the given grammar: $T \rightarrow FT'$ $T' \rightarrow * FT'_1$ $T' \rightarrow \epsilon$ $F \rightarrow digit$ Further generate Annotated parse tree for 3* 5	10	CO1 / CO4	U, E
6.	Considering the following Grammars and the semantic rules generate DAG for the expression $a * (b - c) + (b - c) * d + d$	10	CO5	C, E

	PRODUCTION	SEMANTIC RULES
1)	$E \rightarrow E_1 + T$	$E.node = \textbf{new } Node('+', E_1.node, T.node)$
2)	$E \rightarrow E_1 - T$	$E.node = \textbf{new } Node('-', E_1.node, T.node)$
3)	$E \rightarrow T$	$E.node = T.node$
4)	$T \rightarrow (E)$	$T.node = E.node$
5)	$T \rightarrow \textbf{id}$	$T.node = \textbf{new } Leaf(\textbf{id}, \textbf{id}.entry)$
6)	$T \rightarrow \textbf{num}$	$T.node = \textbf{new } Leaf(\textbf{num}, \textbf{num}.val)$

Generate Three Address Code for the same and represent in Quadruples and Triples format.

OR

State pumping-lemma for Regular Language. Justify whether the given languages are regular or not.

i) $L = \{1^{n^2} : n \geq 0\}$

ii) $L = \{w \in (0 + 1)^*$: numbers of 0s are not equal to numbers of 1 in $w\}$.

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

END SEMESTER EXAMINATION – May, 2024

B. Tech (Computer Science and Engineering) 4th Semester (Section 1, 2, and 3)

***Course: Automata Theory and Compiler Design* (CS44114)**

Full marks:60 Time: 3 hours

Answer *all* questions

Q.no	Question	Marks	CO	BL
1	a) Design Deterministic PDA for the language $L_1 = \{a^m b^n c^{m+n}, m > n \text{ and } n >= 1\}$ b) Define and explain with an example about the synthesized attributes and inherited attributes.	06M 06M	CO4 CO6	Application Remembering & Understanding
2	Consider the following grammar: S→Aa|bAc|Bc|bBa A→d B→d Compute the sets of LR(0) items and show them using the LR(0) automaton (goto graph). Construct the corresponding SLR parsing table and verify whether the given grammar is SLR(1) or not.	12M	CO3	Understanding & Application
3	Design the SDD to convert the given infix expression into postfix expression using the following grammar: E→E+T | T T→T*F | F F→(E) | id Also construct the annotated parse tree for the string *id+id*id*.	12M	CO6	Creation
4	a) Generate the three-address codes for the following program segments and also construct the basic blocks. **int** k = 1; **int** a[5]; **while** (k<= 5) { a[k] = 0; k++; } b) Generate the three-address codes for the following program segment and represent the code in Quadruple and Triples. ***for(i = 1; i<=10; i++) { a[i] = x * 5; }***	06M 06M	CO7 CO7	Understanding & Analyze Understanding & Analyze

5	(a) Translate the expression (a * b / c) * (d + e +f) into three address codes. Write the code (target language instructions using assembly language) by using the three address codes generated above. Also, show the register descriptor list, address descriptor list after each and every instruction generated and the cost of each instruction and total cost of the code.
	(or)
	(b.i) Construct the basic blocks and the corresponding flow graph for the following program fragment:
	x=1;
	for (n=1; n<=3; n++)
	x=x*2;
	(b.ii) Design a Turing Machine for the following language:
	$L=\{WW^R \mid W \in \{0,1\}^*\}$
		12M
		(6+6)
			CO7
			CO7
			CO8
				Application
				Application
				Creation